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Pore-scale numerical investigation of the impacts of surface roughness: Upscaling of reaction rates in rough fractures

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Abstract

The roughness of solid surfaces influences mineral dissolution rates by affecting flow and transport in the near-surface regions and by increasing the surface area available for reaction. The impact of surface area is commonly accounted for by using the surface roughness factor (SRF), which is the ratio between the total surface area and the nominal or geometric surface area. The coupled impacts of hydrodynamics and transport, however, are rarely considered. In this study, we performed pore-scale reactive transport simulations in a series of synthetic 2D rough fractures to investigate the compound effects of surface roughness on the reaction rates in fractures. Simulation results show that while reaction rates increase with SRF, the increase is not linearly proportional to that of the surface area. Rather, local concentration gradients resulting from flow and transport processes limit the increase in the rate. In addition, surface roughness gives rise to concentration gradients that do not otherwise develop in the flat-surface geometries typically considered in modeling studies. To describe the impacts of the surface area increase on reaction rate at different roughness and flow velocities, three distinct regimes were identified. A unified mathematical relationship was also developed that allows the reaction rate in a rough fracture to be approximated by the well-mixed reactor reaction rate and a correction factor. The correction factor follows a power-law function of SRF, with the multiplying factor and exponent expressed as exponential functions of the Péclet and Damköhler number. This mathematical formulation provides a valuable upscaling approach for effective integration of sub-grid scale surface roughness in larger scale continuum models.

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Keywords: Surface roughness; Reaction rate; Upscaling; Fracture; Pore-scale reactive transport model

1. INTRODUCTION

Reaction rates measured in the lab are typically orders of magnitude higher than those observed in the field. This discrepancy has been attributed to two types of factors: intrinsic and extrinsic (White and Brantley, 2003; Liu et al., 2015). The intrinsic factors include the density and diversity of reactive sites (Gautier et al., 2001; Maher et al., 2006; Fischer et al., 2014), while extrinsic factors include hydrological and microbial processes (Maher, 2010; Wen and Li, 2017). For example, the residence time in the field is typically longer and the reactions are close to equilibrium compared to lab experiments (Maher, 2010). Passivation due to the development of a weathered layer or secondary precipitation observed in the field may have also contributed to the lower reaction rate (Noiriel et al., 2007; Maher et al., 2009; Navarre-Sitchler et al., 2011). Some investigators have suggested that the scale dependence of reaction rates contributes to this discrepancy (Li et al., 2008). Given the dependence of reaction rates on scales, a fractal description was proposed to predict the reaction rate of basalt weathering at a given length scale

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from BET scale reaction rate (Navarre-Sitchler and Brantley, 2007). Overall, this scale dependence is a manifestation of concentration gradients resulting from heterogeneity of flow, transport and reactions operating at the fine scale. This observation also highlights the challenge for large-scale modeling, in which homogenization of fine scale heterogeneity is a common practice. Therefore, mechanistic understanding of how fine scale heterogeneity affects large scale processes and properties is needed for the resolution of scientific conundrums such as the scale dependence of reaction rates, and for the development of modeling approaches that allow effective upscaling of fine-scale heterogeneity.

One type of heterogeneity that is of general interest and is the focus of this study is surface roughness. Surface roughness affects the reactivity of the solid phase by increasing surface area. As shown in the commonly used reaction rate laws, reactivity of the solid phase is typically treated as linearly proportional to the physical surface area. Quantification of surface roughness, however, is challenging because of the scale dependence of surface measurements and the large number of metrics that have been used to define surface roughness in different disciplines, including height difference, root mean square roughness, joint roughness coefficient, surface roughness factor, and fractal dimension and amplitudes (Fischer and Luttge, 2007; Bandstra and Brantley, 2015). The selection of surface roughness measurement should be guided by the nature of the studies. Traditionally in geochemistry, a surface roughness factor (SRF) is used to account for the impacts of surface roughness on reaction rates (Fletcher and Brantley, 2010; Beckingham et al., 2016). SRF is defined as the ratio between the total surface area, which is typically determined by the Brunauer-Emmett-Teller method, and the geometric surface area, which can be estimated based on ideal geometry or computed tomography images (Peters, 2009; Lai et al., 2015).

Surface roughness can also influence the reaction rate by affecting flow and transport. For example, surface roughness can increase the hydraulic tortuosity in the fracture flow and result in fracture permeability decrease (Deng et al., 2013; Jin et al., 2017). The effective permeability has been reported to have a power law dependence on the mean fracture aperture, and the exponent of the power law relationship is determined by surface roughness that is characterized by the Hurst exponent (Jin et al., 2017). It has also been observed that compared to a smooth fracture of the same average aperture, surface roughness leads to an earlier breakthrough of a conservative solute and a longer tailing. The early breakthrough is ascribed to higher velocity in the middle of the rough fracture, whereas the long tailing is a result of the development of the relatively immobile zones in the roughness (Bolster et al., 2009, 2014; Cardenas et al., 2009; Bouquain et al., 2012; Chaudhary et al., 2013; Sund et al., 2015). Consequently, the dispersivity coefficients are functions of not only the Péclet number, but also the fracture geometry and the Reynolds number (Bouquain et al., 2012; Richmond et al., 2013). Péclet number measures the relative magnitude of advection and diffusion, and is defined by the average velocity ($\bar{u}$), the average aperture (b), and the molecular diffusion coefficient (D_m).

$$Pe = \bar{u}\, b/D_m \tag{1}$$

Reynolds number is the ratio between the inertial force and the viscous force,

$$Re = \bar{u}\, L/v \tag{2}$$

where L and v are the fracture length and the kinematic viscosity of the fluid, respectively.

Accounting for the impacts of surface roughness on reaction rates over a rough surface is challenging not only because roughness affects reaction rate through surface area, hydrodynamics and transport, but also because solid phase reactivity, hydrodynamics and transport are intimately coupled. As such, a scaling factor, in addition to the surface roughness factor, has typically been used to correct surface area for mineral accessibility or transport limitation (Moore et al., 2012; Beckingham et al., 2017). The determination of the scaling factor, however, is arbitrary and is often a calibration exercise rather than having a physical basis. Few studies have investigated the interplay between flow, transport and reaction in rough fractures or pores and the subsequent impacts on mass transfer of a reacting solute or bulk reaction rate. These studies further highlighted the need to account for the compound effects of surface roughness on reaction rates in a rough fracture or pore. Jeschke and Dreybrodt (2002) investigated the extent to which surface area associated with micro-pores and complex surface morphology contributes to reactions under conditions commonly used for experimental determination of mineral reaction rates. Using a linear reaction rate, the authors developed analytical solutions for the transport equation in presence of a diffusion boundary layer. While the solutions allow quantification of the contribution of surface area of open pores to reactions, they do not account for the development and dynamic change of the diffusive boundary layer as a result of surface morphology, reaction and flow. Dykaar and Kitanidis (1996) studied the relationship between the micro-scale and macro-scale reaction and transport properties for a reacting solute that is being consumed at a linear rate in a porous medium, which was represented by a sinusoidal structure. Their numerical results showed that the relationship varies with the Péclet and Damköhler number (which measures the relative magnitude between the time scales of flow or transport and reaction) of the system. Sund et al. (2015) investigated the transport of a reacting solute through a rough channel and demonstrated that an upscaled random walk model can be used to capture the breakthrough curve of the reacting solute. Cvetkovic and Gotovac (2014) upscaled chemical transport in fracture networks using a time domain random walk approach by incorporating heterogeneity of the flow in the effective surface area. These studies, however, focus on linear reaction systems, and it has been suggested that further investigations are needed for nonlinear reactions (Sund et al., 2015).

Therefore, there is a research need to evaluate and quantify the compound effects of surface roughness on nonlinear water-rock reactions. The objectives of our study are to

provide mechanistic understanding of how surface roughness affects reaction rates in rough fractures, and ultimately to develop an upscaling approach that integrates the compound effects for large scale simulations. To this end, we performed pore-scale reactive transport simulations in a series of idealized and realistic rough fractures with different levels of roughness. Fractures, as preferential flow conduits, are of importance in many geological systems. Understanding mineral reaction rates in fractures is important for the prediction of the dynamic evolution of fractures. As this study primarily focuses on the interplay between reaction, flow and transport over rough surfaces, no intrinsic heterogeneity in reactivity was considered, i.e. the reaction kinetics were assumed to be the same regardless of the roughness. Pore-scale reactive transport simulation results were also compared with continuum-scale models with different levels of simplifications so as to isolate different factors that can affect reaction rates. Details of geometry generation, the pore-scale reactive transport model, and the set-up of the numerical simulations are provided in Section 2. In Section 3, we present detailed flow and concentration fields from the full roughness simulations, comparisons of different simulations and a mathematical relationship for upscaling. We conclude with a discussion and summary of the findings.

2. METHODS

2.1. Geometry generation

A series of 2D rough profiles were used for the simulations. The 2D profiles represent cross-sections perpendicular to the fracture plane (Fig. 1) and can be considered as 3D fractures of unit width with parallel profiles in the direction perpendicular to the flow. All fracture cross-sections have a length (L) of 1000 μm and an average aperture (b) of 100 μm. The size is typical of a micro crack, a single pore, or a single grid cell in a larger fracture domain (Li et al., 2008; Deng et al., 2017). Even though the flow velocity profile between two plates of a fracture is different from that in a 2D cross-section of a cylindrical pore, the observations for fractures are extendable to single pores (Li et al., 2008).

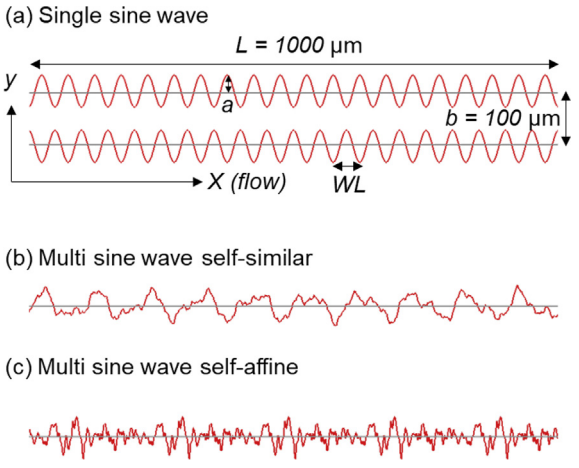


Fig. 1. (a) A rough fracture generated using a single sine wave with a wavelength of 50 μm and an amplitude of 30 μm. (b) A self-similar rough profile (lower half of the fracture) generated by overlaying 20 sine waves with wavelengths ranging from 0.0005 to 100 μm, and an amplitude-to-wavelength ratio of 0.2. (c) A self-affine rough profile (lower half of the fracture) generated by overlaying 43 sine waves with wavelengths ranging from 4.35 to 50 μm, and an amplitude-to-wavelength ratio of 0.2.

A fracture cross-section includes a rough surface and its mirror symmetry. The rough surfaces were created using sine waves. Three types of rough profiles were considered (Fig. 1).

The first type of rough profiles represents idealized surface roughness and involves a single sine wave (Eq. (3)). Regardless of its simplicity, this type of rough profiles has been widely adopted and proven to provide useful insights regarding fluid flow and solute transport in rough fractures (Bolster et al., 2009; Bouquain et al., 2012; Richmond et al., 2013; Sund et al., 2015). Fourteen rough profiles (Table 1) were generated using a single sine wave with wavelengths (WL) that range from 10 to 100 μm and amplitudes (a) that are of 0.1 ~ 1.5 times of the wavelength. For these profiles, the phase of the sine waves (x_0) is half of the wavelength.

$$y = a \cdot \sin\left(\frac{2\pi}{WL}(x - x_0)\right) \tag{3}$$

The other two types of rough profiles are fractal surfaces. The roughness of fractures and fault surfaces observed in the field and in the lab has also been described by fractal models (Power and Tullis, 1991; Meakin, 1993; Schmittbuhl et al., 1993; Glover et al., 1998; Renard et al., 2013). This means that the measurement of surface features (m_l) follows a power law relationship with respect to the length scale (l) of the measurement (Eq. (4)).

$$m_l \propto l^{N-D} \tag{4}$$

where D is the fractal dimension in an N-dimension space. For fractal surfaces, the subsections are statistically identical to the entire profile when magnified by a scaling factor in each direction. The profiles are self-similar if the scaling factors are the same, and are self-affine if the values are different. For self-similar rough profiles,D is close to one. For self-affine rough profiles, the Hurst exponent (H), also known as the roughness exponent, is used to measure how the surface roughness changes with scale. It has a value between zero and one, and is related to the fractal dimension by $D = 2 - H$ for a 2D profile. Both self-similar and self-affine fractures have been documented (Power et al., 1987; Schmittbuhl et al., 1993; Renard et al., 2013). For natural self-affine fractures in geological systems of different scales, the Hurst exponent has been reported to converge to 0.8 (Schmittbuhl et al., 1993; Renard et al., 2013).

In our study, the fractal geometries were synthesized by overlaying a series of sine waves with the same amplitude-to-wavelength ratio and random phases (Eq. (5)), following Power and Tullis (1991).

$$y = \sum\nolimits_i a_i \cdot \sin(f_i(x - x_{i,0})) \tag{5}$$

where f_i is the frequency of the sine wave, i.e. $\frac{2\pi}{WL_i}$. If the frequencies of the sine waves follow a geometric progression (Eq. (6)), the generated rough profiles are self-similar.

Table 1
Geometric statistics of the rough profiles. WL – wavelength [μm], WLmax – maximum wavelength [μm], WLmin – minimum wavelength [μm], a/WL – amplitude-to-wavelength ratio, SRF – surface roughness factor, #W – number of sine waves, H – Hurst exponent.

	WL	a/WL	SRF	#W	WL_{max}	WL_{min}	a/WL	SRF		#W	WL_{max}	WL_{min}	a/WL	SRF	H
Geo1			1.00	Geo16	17	250	0.3806	0.05	1.27	Geo34	74	500	3.88	0.05	1.93	0.78
Geo2	100	0.2	1.09	Geo17	18	250	0.0302	0.05	1.22	Geo35	80	250	8.16	0.05	2.00	0.80
Geo3	100	0.4	1.32	Geo18	18	250	0.0302	0.1	1.66	Geo36	72	100	8.58	0.05	1.90	0.79
Geo4	100	0.6	1.62	Geo19	19	250	0.0178	0.05	1.22	Geo37	72	100	8.58	0.1	3.24	0.79
Geo5	100	0.8	1.95	Geo20	19	250	0.0178	0.1	1.66	Geo38	68	100	5.63	0.05	1.86	0.79
Geo6	50	0.4	1.32	Geo21	17	100	1.5028	0.05	1.30	Geo39	68	100	5.63	0.1	3.15	0.79
Geo7	50	0.8	1.95	Geo22	17	100	1.5028	0.1	1.86	Geo40	43	50	4.35	0.15	3.81	0.79
Geo8	50	1	2.30	Geo23	18	100	0.0339	0.05	1.21	Geo41	43	50	4.35	0.2	4.97	0.79
Geo9	50	1.2	2.67	Geo24	18	100	0.0339	0.1	1.64	Geo42	45	50	7.69	0.15	3.71	0.83
Geo10	20	1	2.30	Geo25	20	100	0.0132	0.05	1.21	Geo43	52	50	3.64	0.1	2.86	0.77
Geo11	20	2	4.19	Geo26	20	100	0.0132	0.1	1.63	Geo44	52	50	3.64	0.15	4.07	0.77
Geo12	20	3	6.14	Geo27	20	100	0.0132	0.2	2.63	Geo45	52	50	3.64	0.2	5.30	0.77
Geo13	10	1	2.30	Geo28	17	100	0.0035	0.05	1.16							
Geo14	10	2	4.18	Geo29	17	100	0.0035	0.1	1.51							
Geo15	10	3	6.13	Geo30	17	100	0.0035	0.2	2.37							
				Geo31	20	100	0.0005	0.05	1.16							
				Geo32	20	100	0.0005	0.1	1.51							
				Geo33	20	100	0.0005	0.2	2.37							

$$f_i = \frac{2\pi}{WL_{max}} r_f^{i-1} \tag{6}$$

If the frequencies instead follow an arithmetic sequence (Eq. (7)), the generated rough profiles are self-affine.

$$f_i = \frac{2\pi}{WL_{max}} + (i-1)d \tag{7}$$

Parameters that control the rough profiles, including the maximum wavelength, the amplitude-to-wavelength ratio, the phase of each sine wave ($x_{i,0}$), the ratio (r_f) of the geometric sequence of frequencies in the self-similar case and the difference (d) of the arithmetic sequence in the self-affine case, were generated randomly. Eighteen self-similar and twelve self-affine rough profiles were selected for the subsequent simulations (Table 1). To ensure that the fracture is open, only profiles for which $y_{max} < \frac{b}{2}$ were included. For the self-affine profiles, the Hurst exponent was calculated from the Fourier power spectrum ($F_{ps}(w_n)$) of the rough profiles (Eq. (8)) (Renard et al., 2004).

$$F_{ps}(w_n)^2 \propto w_n^{-1-2H} \tag{8}$$

where w_n is the wave number of the power spectrum. The selected self-affine profiles all have a Hurst exponent (H) of about 0.8.

While the generation of the rough profiles was constrained by the sine wave parameters and fractal statistics, roughness is measured by the surface roughness factor (SRF) in the following analyses. SRF is defined as the ratio between the total surface area (A_{total}) and the geometric or nominal surface area (A_{geo}) (Eq. (9)).

$$SRF = \frac{A_{total}}{A_{geo}} = \frac{L_{sinewave}}{L} \tag{9}$$

SRF was chosen for the analysis of reaction rate because it has been widely used to account for the impacts of surface roughness, surface area increase associated with roughness in particular, on the reaction rate over rough surfaces, R (Eq. (10)).

$$R = SRF \cdot A_{geo} \cdot k_{rxn} \cdot \left(1 - \left(\frac{IAP}{K_{eq}}\right)^m\right)^n \tag{10}$$

where k_{rxn} is the kinetic coefficient, IAP denotes the ionic activity product, K_{eq} is the equilibrium constant, and the coefficients m and n are typically assumed to be one.

For the 2D cross-sections, A_{total} is the equivalent of the arc length of the sine curves ($L_{sinewave}$), and A_{geo} is the fracture length. $L_{sinewave}$ can be calculated directly from the sine wave functions. The analytical solutions (Eqs. (11b) and (11c)) derived by substituting Eqs. (3) or (5) into Eq. (11a) show that SRF is closely related to the parameters of the sine functions.

$$L_{sinewave} = \int_0^L \sqrt{1 + (y'(x))^2}\, dx \tag{11a}$$

$$L_{sinewave} = \int_0^L \sqrt{1 + \left(\frac{2\pi a}{WL}\cos\left(\frac{2\pi}{WL}(x - x_0)\right)\right)^2}\, dx \tag{11b}$$

$$L_{sinewave} = \int_0^L \sqrt{1 + \left(\sum_i \frac{2\pi a_i}{WL_i}\cos\left(\frac{2\pi}{WL_i}(x - x_0)\right)\right)^2}\, dx \tag{11c}$$

For the single sine wave geometry, which is not a fractal surface, SRF is dependent on the amplitude and wavelength of the sine wave. For the fractal surfaces, while the fractal dimension or Hurst exponent is effectively the same, SRF provides a single parameter that captures the variations of the complex sine functions.

2.2. Pore-scale reactive transport modeling

The governing equations include the incompressible Navier-Stokes equation for constant density fluid flow (Eq. (12a)) and the continuity equation (Eq. (12b)), and the mass balance equation for multi-component reactive transport (Eqs. (13a) and (13b)).

$$\frac{\partial u}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + v\nabla^2 \boldsymbol{u} \tag{12a}$$

$$\nabla \cdot \boldsymbol{u} = 0 \tag{12b}$$

where $\boldsymbol{u}$ is fluid velocity, ∇p is the pressure gradient, and v is the kinematic viscosity.

$$\frac{\partial C_i}{\partial t} + (\boldsymbol{u} \cdot \nabla) C_i = \nabla \cdot D_m \nabla C_i \tag{13a}$$

where C_i is the concentration of component i, and D_m is the molecular diffusion coefficient. Because reactive surfaces are explicitly considered in a pore scale model, the mineral reaction rate can be expressed as the boundary condition at the fluid-solid interfaces

$$-D_m \nabla \mathrm{C_i} = R_i \tag{13b}$$

where R_i is the surface rate contribution of mineral reactions to the mass of component i, and the surface area is considered directly in the discretized form of the mass balance equation (Molins et al., 2012).

Chombo-Crunch (Molins et al., 2012; Trebotich et al., 2014), a high resolution pore-scale reactive transport model, was used to numerically solve the governing equations. Chombo-Crunch uses a finite volume discretization and an embedded boundary method (Trebotich and Graves, 2015). The solid-fluid interface intersects the otherwise regular Cartesian grid cells, creating the "irregular" grid cells with partial volumes occupied by the fluid and the solid. This method allows complex geometries to be represented rigorously. The embedded boundary approach also improves the computational efficiency in geometrically complex domains because the "irregular" cells are limited to a small fraction of the computational domain.

Chombo-Crunch has been previously used to successfully simulate a capillary experiment, in which CO_2-acidifed fluid was flown through a capillary tube packed with calcite grains (Molins et al., 2014). The same numerical schemes were also used in this study.

2.3. Geochemistry

To focus on surface roughness, the solid phase is composed of a single mineral, calcite. The influent is a 0.01 M NaCl brine acidified by CO_2 under ambient pressure conditions, and has a pH of 5. The aqueous reactions are assumed to reach equilibrium instantaneously. The reactions and equilibrium constants are reported in Table 2. The mineral reaction, i.e. calcite dissolution is treated kinetically using the transition state theory rate law. The kinetic coefficient is calculated based on the three elementary reaction pathways (Chou et al., 1989) (Eq. (14)). The kinetic coefficients of each pathway is summarized in Table 3.

Table 2
Aqueous reactions included in the simulations, and the corresponding equilibrium constants.

Aqueous complexation reaction	LogK_{eq}
$OH^- + H^+ \rightleftharpoons H_2O$	13.99
$CO_3^{2-} + 2H^+ \rightleftharpoons H_2CO_3^*(aq)$	16.67
$HCO_3^- + H^+ \rightleftharpoons H_2CO_3^*(aq)$	6.34
$CaCO_3(aq) + 2H^+ \rightleftharpoons H_2CO_3^*(aq) + Ca^{2+}$	13.35
$CaHCO_3^- + H^+ \rightleftharpoons H_2CO_3^*(aq) + Ca^{2+}$	5.30
$CaOH^+ + H^+ \rightleftharpoons Ca^{2+} + H_2O$	12.85
$CaCl^+ \rightleftharpoons Ca^{2+} + Cl^-$	0.7
$CaCl_2(aq) \rightleftharpoons Ca^{2+} + 2Cl^-$	0.65
$NaCO_3^- + 2H^+ \rightleftharpoons H_2CO_3^*(aq) + Na^{2+}$	16.16
$NaCl(aq) \rightleftharpoons Na^+ + Cl^-$	0.78
$NaHCO_3(aq) + H^+ \rightleftharpoons H_2CO_3^*(aq) + Na^+$	6.18

Table 3
Calcite dissolution reaction and the equilibrium constant, and the three reaction pathways and the corresponding kinetic coefficient (Chou et al., 1989).

Dissolution reaction	LogK_{eq}
$CaCO_3(s) + 2H^+ = Ca^{2+} + H_2CO_3^*(aq)$	8.16
Reactions pathway	Log$k$ $(mol/m^2 s)$
$CaCO_3(s) + H^+ \overset{k1}{\Longleftrightarrow} Ca^{2+} + HCO_3^-$	−0.05
$CaCO_3(s) + H_2CO_3^* \overset{k2}{\Longleftrightarrow} Ca^{2+} + 2HCO_3^-$	−3.3
$CaCO_3(s) \overset{k3}{\Longleftrightarrow} Ca^{2+} + CO_3^{2-}$	−6.19

$$k_{rxn} = k_1 a_{H^+} + k_2 a_{H_2CO_3^*} + k_3 \tag{14}$$

2.4. Simulations

The forty-four rough fracture geometries were simulated in Chombo-Crunch at a range of average flow velocities ($\bar{u}$). All rough profiles were simulated at velocities of 0.0001, 0.001, 0.01, 0.1, 1 and 10 cm/s, and a subsection of the rough profiles were also simulated at 0.0005, 0.005, 0.05, 0.5, 5 and 50 cm/s. The velocity range is within what has been observed in natural and engineered systems (Li et al., 2008), and corresponds to a Reynolds number of 1×10^{-3} to 500. For $\bar{u} \leq 1$ cm/s, the flow is in the Stokes flow regime, where the viscous force dominates; at higher velocities, the flow is still laminar but the contribution of the inertial force becomes significant.

The Chombo-Crunch simulations, also referred to as roughness simulations, explicitly account for the impacts of surface roughness on surface area, longitudinal and transverse diffusion, and local hydrodynamics, and therefore can be considered as the ground truth. Because the fracture was assumed to be symmetric, only half of the fracture domain was simulated. The sine wave functions were

inputted directly into Chombo-Crunch to create the computational domain. The grid cell size is 0.4 μm by 0.4 μm, which is comparable to the resolution of synchrotron-based micro-tomography images and the scale of roughness that can be captured by such images. Roughness above the grid cell size was fully captured. The surface area calculated in Chombo-Crunch instead of the values derived from the sine wave functions was used for the calculation of SRF so as to avoid over-counting of the roughness that is below the size of the grid cells used in the simulations.

A set of pore-scale reference simulations was performed with a domain equivalent to that of the roughness simulations – i.e. with an aperture of 100 μm – but with flat surfaces (Fig. 2b). The difference between reaction rates from this set of simulations and the roughness simulations at the same velocity is due to the compound effects of roughness.

Three additional sets of simulations were performed (Fig. 2) to isolate individual factors. In the well-mixed reactor simulations, the fracture is represented as a single grid cell where sub-grid flow and transport are not considered. Longitudinal diffusion is explicitly considered in the 1D simulations, whereas the 2D simulations account for both longitudinal and transverse diffusion.

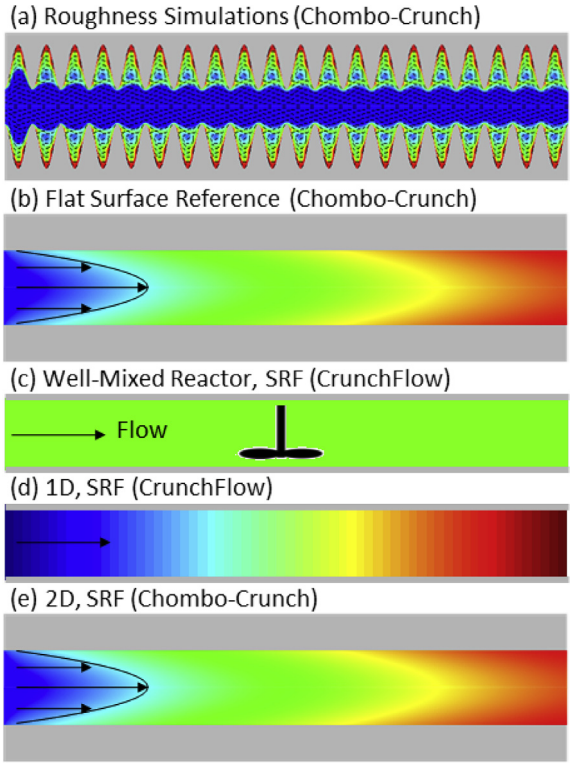


Fig. 2. Conceptual illustration of the different modeling approach and codes used in the simulations. (a) Roughness simulations using Chombo-Crunch, (b) a 2D flat-surface reference, (c) a well-mixed reactor, (d) a 1D model, and (e) a 2D model. Roughness in (a) and (b) is captured explicitly in the simulations, while SRF is used in (c)–(e). The color-map illustrates cation concentration with the warmer color corresponds to a higher concentration and the arrows indicate flow.

In these three sets of simulations, unlike the roughness simulations for which the rough geometry is resolved explicitly, surface roughness is simply considered by correcting for the increase in surface area using the SRF. A range of SRF values (1–6) were used in the simulations so that the resulting reactive surface area covers the range encountered in the roughness simulations. For each model, the simulations with SRF of one serve as the smooth fracture reference.

The well-mixed reactor and the 1D simulations were performed using CrunchFlow, a continuum reactive transport code (Steefel et al., 2015). In the 1D simulations, the fracture was discretized along the flow direction into 1000 grid cells. The 2D simulations were performed in Chombo-Crunch, using the same domain size and grid cell size as the roughness simulations. In the 2D simulations, although surface area increases with SRF, the geometry is the same as the flat-surface reference simulation noted earlier. Thus, local hydrodynamics that arise from roughness are not captured.

The bulk reaction rate of the computational domain was calculated from the average calcium (Ca) concentration at the outlet ($C_{Ca,out}$) and the volumetric flux (Q). The reaction rate was not normalized against surface area to avoid making assumptions about the surface area that contributes to the reactions.

$$R = C_{Ca,out} \cdot Q \tag{15}$$

For the 2D and roughness simulations, both Ca concentration and fluid flux (q) vary across the fracture aperture. The average Ca concentration used in Eq. (15) is the concentration weighted by the flux.

$$C_{Ca,out} = \frac{\int_0^{b/2} C_{Ca}(y) q(y)}{Q} \tag{16}$$

2.5. Upscaling

Typically, a micro-crack or a grid cell in a numerical model of a larger fractured domain is represented by a well-mixed reactor, for which roughness cannot be resolved at the fine scale. For upscaling, it is necessary to develop a mathematical relationship between the rough-fracture reaction rate (R_{rough}), which includes the impacts of surface roughness, and the rate in a flat fracture as predicted by a well-mixed reactor model ($R_{WMR,flat}$). A general form is given in Eq. (17), where F is a correction function.

$$R_{rough} = F R_{WMR,flat} \tag{17}$$

Non-linear analysis was performed using the reaction rates simulated from the pore-scale model to formulate F as a function of SRF, which measures surface area increase, and Péclet number (Pe) and Damköhler number (Da_I), which provide measures of the hydrodynamics and their interplay with reactions. Péclet number (Pe) was defined in Eq. (1), and Damköhler number (Da_I) can be calculated from Eq. (18),

$$Da_I = k_{rxn} L / \left(1000 b \bar{u} C_{eq}\right) \tag{18}$$

where k_{rxn} is the kinetic coefficient that is determined by the influent composition (Eq. (14)), C_{eq} is the equilibrium concentration, and the factor 1000 is used to convert the unit of liter to m^3. This formulation compares the advection time scale with the time scale that is needed for the inflowing reactive fluid to reach equilibrium with respect to the mineral phase. In this case, C_{eq} is the Ca concentration of a batch system where the influent reaches equilibrium with respect to the mineral phase, and has a value of 0.16 mM.

For the non-linear analysis, 2528 data points from the pore-scale reactive transport simulations were used. These data points include the full-length fracture simulation results, as well as results that represent a variety of shorter fractures. In addition to the fracture outlet, flux-weighted Ca concentrations were sampled at nine different lengths within the fracture domain that are evenly distributed at an interval of 100 μm. The concentrations were used to calculate the bulk reaction rate for the fracture segment between the inlet and the sampling location. Da_I of the corresponding shorter fractures was also calculated using Eq. (18). Some data points of the shorter fractures were excluded from the analysis when the corresponding sampling points are located in deep troughs, and the flow fields or solute concentrations close to the sampling points are obviously affected by the downstream geometry. In these cases, the boundary condition is not comparable to the full-length fracture simulations.

3. RESULTS

3.1. The impacts of surface roughness

Here, the ratio between the reaction rate of a rough fracture and the reaction rate of the reference flat fracture (R_{rough}/R_{flat}) is reported as a measure of the impacts of surface roughness on reaction rate. The reaction rates of the reference smooth fracture vary between 1×10^{-12} and 1×10^{-10} mol/cm^2 s across the flow velocities simulated (Fig. 3), and are comparable to the values reported in previous studies under similar conditions (Li et al., 2008; Levenson et al., 2015). The reaction rate increases with flow velocity and asymptotically approaches the surface controlled value. Fig. 4 plots the results from the roughness simulations.

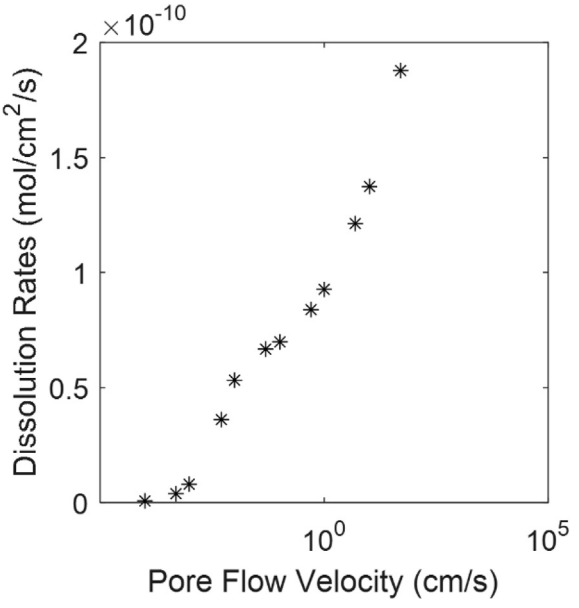


Fig. 3. Calcite dissolution rate of the reference flat fracture. The reaction rate is normalized against the geometric surface area.

The reaction rate ratio is larger than one because reaction rate over the rough surface is elevated by surface area increase. The ratio increases with SRF (Fig. 4(a)), but the increase is not linearly proportional to SRF, which is what would be expected based on the reaction rate law (Eq. (7)). Instead, the reaction rate ratio increases to a lesser extent than it would if proportional to the increase in reactivity expressed by the SRF. For example, with an SRF of ~6, the increase in reaction rate is <4 across the velocity range simulated. This implies that the potential for reaction rate increase due to surface roughness as indicated by the surface area increase is not fully realized, and suggests that factors other than surface area also affect the reaction rates over rough surfaces.

For a given flow velocity, the increase in the reaction rate ratio in relation to SRF shows a consistent trend among the three types of rough profiles (Fig. 4(a)). This observation suggests that despite its simplicity, for the range of velocity and roughness tested, the single sine wave geometry involves similar interactions between reaction, flow and transport that are present in the realistic rough fractures. Although results will be presented for different types of rough profiles in what follows, they will not be discussed individually.

For a given SRF, the reaction rate ratio increases with flow velocity initially, and decreases as the flow velocity increases further, i.e. $\bar{u} \geq 0.1$–0.5 cm/s (Fig. 4(b)). The initial increase indicates the transition from a transport-controlled reaction rate to a surface reaction rate. In the surface reaction rate controlled regime, the reaction rate depends on the surface area. At higher flow velocities, the ratio between the rate for a rough fracture and a flat-fracture decreases again, indicating that the surface area associated with roughness does not contribute to the reactions due to local transport limitations. Previous studies of smooth single fractures (e.g. (Li et al., 2008)) have noted that concentration gradients and the reaction rate regime change as a function of flow rate. As flow velocities increase, concentration gradients first develop and then are reduced because of the thinning of hydrodynamic boundary layer. Here, the impact of roughness on reaction rates at different flow velocities is explored in details.

In the remainder of this section, the roughness simulations are compared with the well-mixed reactor, 1D and 2D simulations, respectively. The results of three profiles (Geo11, Geo33, Geo42) are used as representative of each type of rough geometry in the comparisons. Overall, three distinct regimes are identified based on reaction rate comparisons and examinations of the flow field and concentration profiles in the rough fractures.

Regime I As shown in Fig. 5(a), at the low velocities, the increase in SRF results in either a highly limited increase in

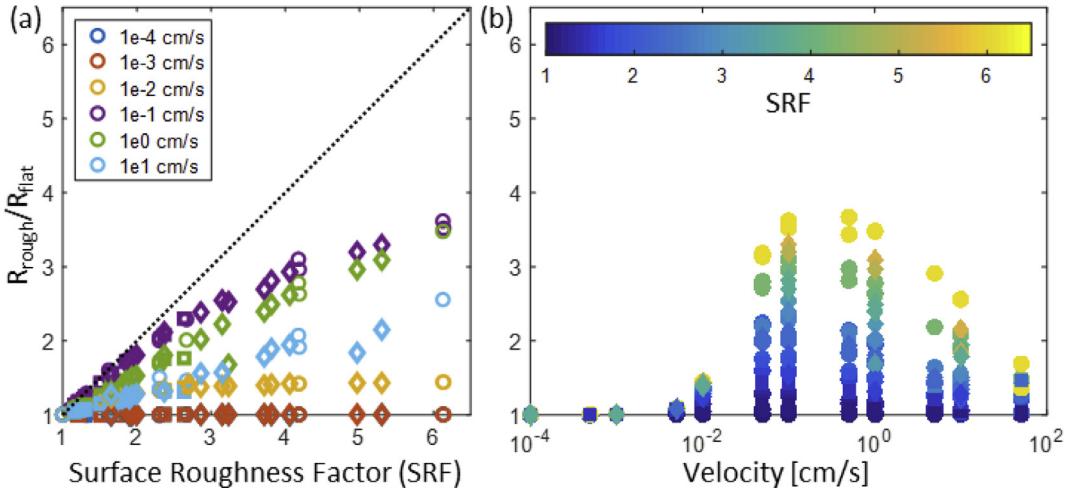


Fig. 4. The ratio between the reaction rate in a rough fracture and the reaction rate in a reference flat fracture in relation to (a) surface roughness factor and (b) flow velocities. The data points (○ – single sine wave profile, □ – self-similar profile, ◇ – self-affine profile) are results predicted by the roughness simulations. The hollow symbols in (a) are color-coded by velocity, and the filled symbols in (b) are color-coded by SRF.

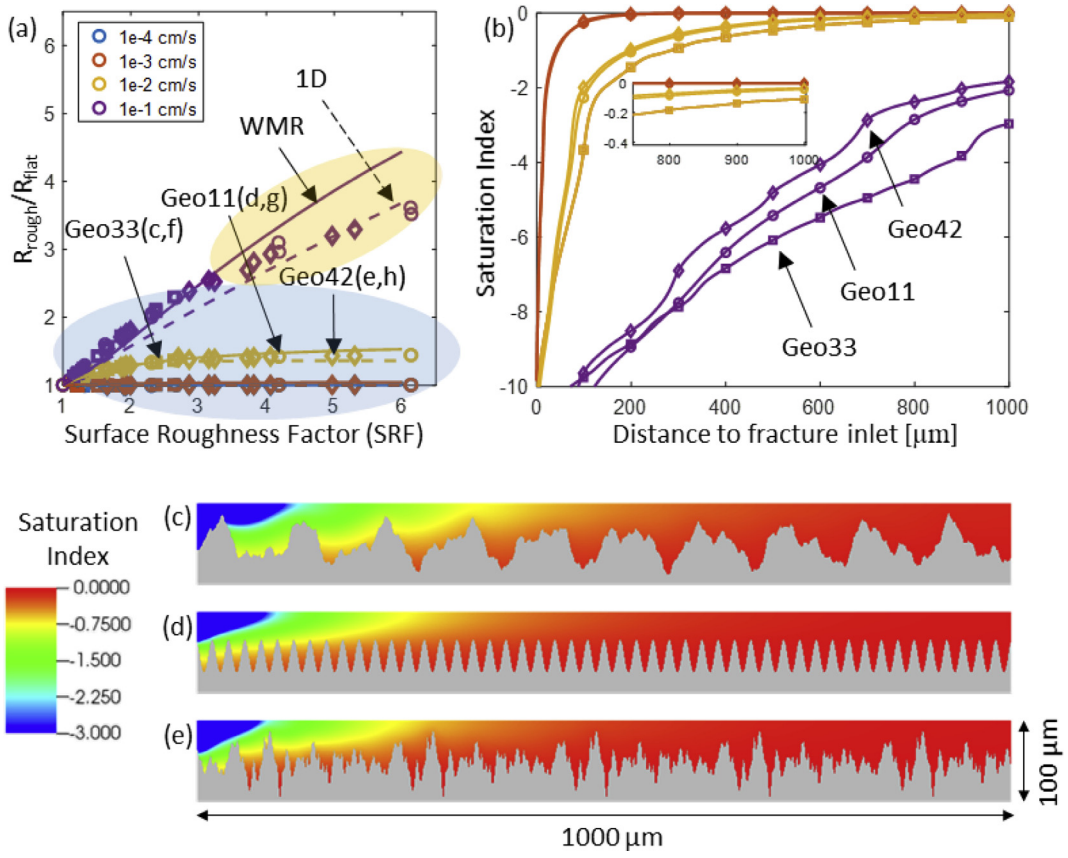


Fig. 5. (a) The ratio between the reaction rate over a rough profile and the rate over a flat profile plotted against surface roughness factor at velocities ranging from 0.0001 to 0.1 cm/s. The data points (○ – single sine wave profile, □ – self-similar profile, ◇ – self-affine profile) are results predicted by the roughness simulations. The solid lines plot the results from the well-mixed reactor, and the dashed line is the 1D results. The blue oval and yellow oval highlight the data points of roughness simulations that agree well with the well-mixed reactor simulations and the 1D simulations, respectively. (b) Saturation index at the center of the fracture for geometric profile 33 (□), 11(○), and 42 (◇) at velocities of 0.001, 0.01, 0.1 cm/s (using the same color-code as (a)). (c–e) Color maps of the saturation index with respect to calcite at 0.01 cm/s in geometric profile 33, 11, and 42, respectively.

the overall reaction rate (e.g. 0.01 cm/s) or no increase at all, such as at the flow rates of 0.0001 and 0.001 cm/s. The reaction rate does not increase with SRF increase when the fluid has reached equilibrium with respect to the mineral phase within the fracture domain (Fig. 5(b)). In cases where the fluid has not equilibrated with the mineral phase when exiting the outlet (e.g. SI is ~−0.1 at 0.01 cm/s, Fig. 5(b)), the overall reaction rate can still increase with surface roughness, i.e. surface area.

The less negative saturation index values throughout the fractures (Fig. 5(c)–(e)), corresponding to high Ca concentrations in the fractures, indicate that fluid in a large portion of the fractures is close to equilibrium. As a result, the thermodynamic driving force is small in a large portion of the fractures, and the local reaction rate does not contribute to the overall reaction rate significantly. In this case, therefore, only in a small portion of the fracture that is close to the inlet does surface area increase due to surface roughness actually contribute to the overall reaction rate. The observations are consistent with previous investigations that showed minor impacts of roughness on the transport of a reacting solute (Yeo, 2001).

The increase in reaction rate due to surface roughness predicted by the roughness simulations is in agreement with the increasing trend predicted by the well-mixed reactor simulations that account for the increase in surface area. As the concentration gradient in the fracture domain is relatively small in these cases (Fig. 5(c)–(e)), the well-mixed assumption applies. This is also consistent with previous observations in parallel-wall fractures that the discrepancy

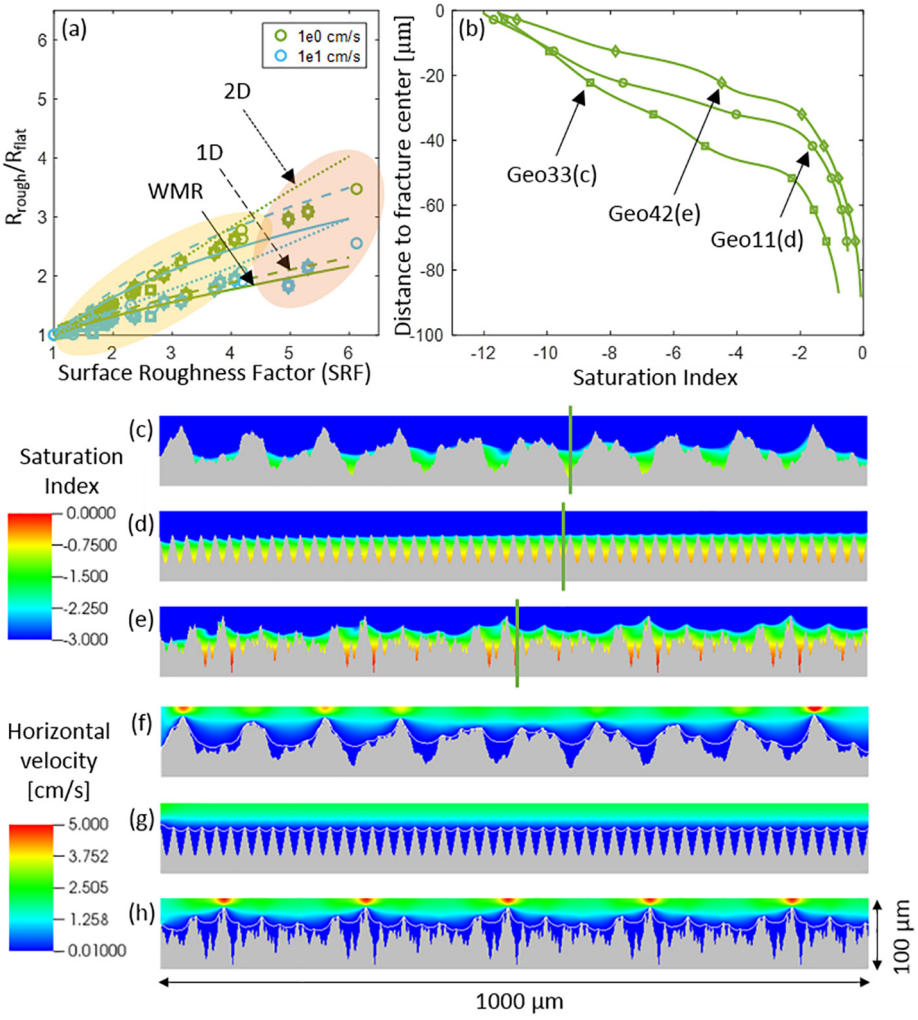


Fig. 6. (a) The ratio between the reaction rate over a rough profile and the rate over a flat profile plotted against surface roughness factor at velocities 1 and 10 cm/s. The data points (○ – single sine wave profile, □ – self-similar profile, ◇ – self-affine profile) are results predicted by the roughness simulations. The solid lines, dashed lines and dotted lines are results from the well-mixed reactor, 1D and 2D simulations, respectively. The yellow oval and orange oval highlight the data points of roughness simulations that agree well with the 1D and 2D simulations, respectively. (b) Saturation index from the center of the fracture to the fracture surface at 1 cm/s for geometric profile 33 (□), 11 (○), and 42 (◇) at the location highlighted in (c–e). (c–e) Color maps of saturation index with respect to calcite at 1 cm/s in geometric profile 33, 11, and 42, respectively. (f–h) Color maps of horizontal velocities at 1 cm/s in geometric profile 33, 11, and 42, respectively. The white lines highlight the diffusion boundary layer where the horizontal velocity is 1% of the average flow velocity.

between reaction rate predicted by a 2D model and that predicted by a well-mixed reactor model is small at low flow velocities (Li et al., 2008). Surface roughness does not change this observation except at higher flow rates, which give rise to Regime II.

Regime II As flow velocity increases, the increase in reaction rate in relation to SRF becomes more significant (Figs. 5(a) and 6(a)). This reflects the fact that with the shorter residence time, the fluid in the fracture is far from equilibrium, and the increase in surface area due to roughness contributes to reactions. The increase, however, can no longer be captured by the well-mixed reactor.

At 0.1 cm/s, the saturation index along the center of the fracture shows a large gradient (Fig. 5(b)). It is also evident that the longitudinal gradient is affected by surface roughness. With larger SRF (e.g. Geo42), the saturation index increases more sharply because of the larger surface area and thus more calcite dissolution at the upstream. In this case, the longitudinal gradient can no longer be ignored. A 1D model that considers the longitudinal diffusion is needed to simulate the increase in the reaction rate ratio with respect to SRF, as highlighted by the yellow oval shade in Fig. 5(a).

At velocities of 1 and 10 cm/s, the concentration gradient across the fracture aperture, which is demonstrated by the gradient of the saturation index, becomes substantial throughout the fracture domain (Fig. 6(b)–(e)). Consequently, transverse diffusion must be included as in the 2D model to capture the reaction rate increase caused by surface roughness. The transverse concentration gradient is a result of the relatively fast surface reaction rate of the mineral phase and the velocity gradient across the fracture aperture (Fig. 6(f) and (g)). A hydrodynamic boundary layer, which is defined by the contour of the horizontal

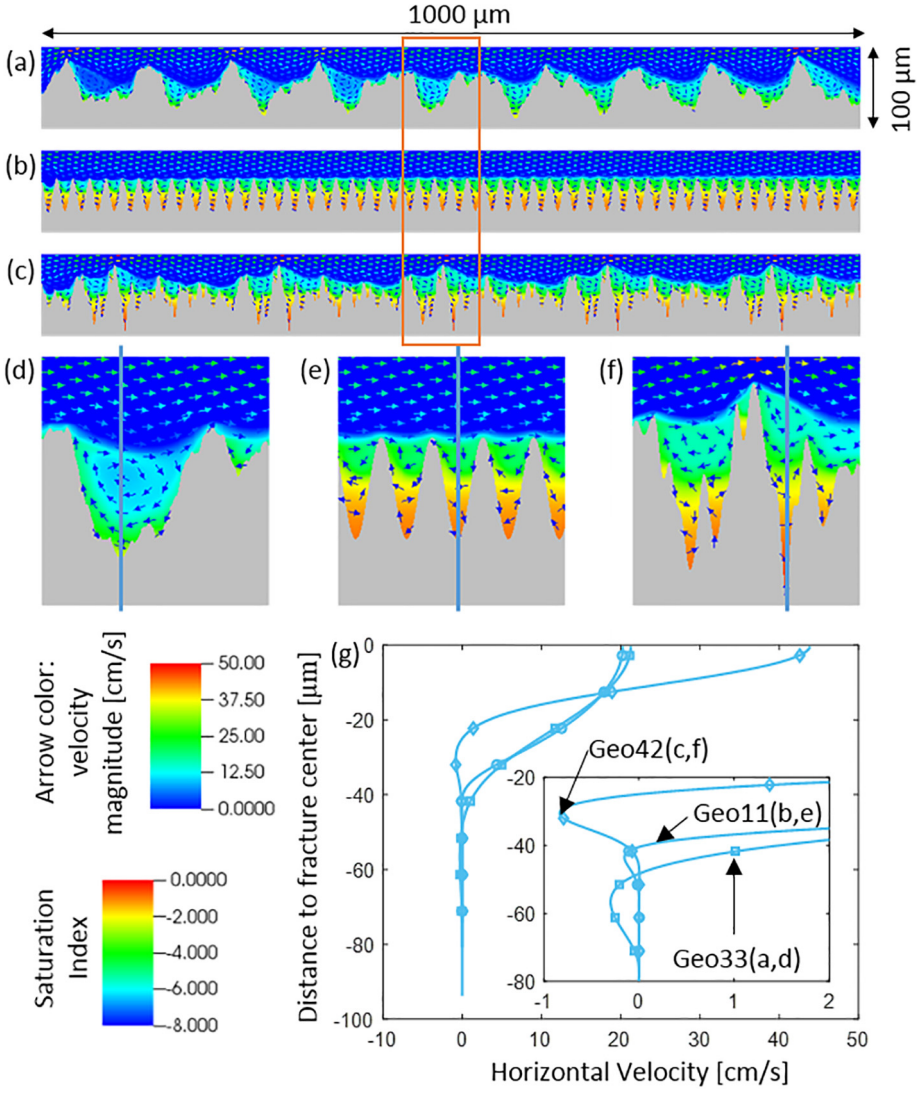


Fig. 7. (a–c) Saturation index and velocity vector at 10 cm/s for geometric profile 33, 11, and 42, respectively. (d–f) Blow-ups of the regions highlighted in orange in (a–c). (g) Horizontal velocity of the cross sections highlighted in (d–f) for geometric profile 33 (□), 11 (○), and 42 (◇). Negative velocity indicates the change of flow direction.

velocity that is 1% of the average flow velocity here, forms, especially in the troughs. The low velocity in the boundary layer implies that mass transfer close to the fluid-solid interface is transport-limited. The resulting concentration gradient controls the local thermodynamic driving force, and thus the increase in local reaction rate in response to surface roughness. For example, the solution in the immobile zones in the troughs is closer to equilibrium, and thus the surface area in the troughs is unlikely to contribute to the overall reaction.

At low roughness, accounting for the transverse concentration gradient and surface roughness factor using the 2D model reproduces the increase in the reaction rate correctly, as highlighted by the yellow oval shade in Fig. 6(a).

Regime III As roughness increases, the 2D simulations over-predict the increase in reaction rate ratio (Fig. 6(a)). As shown in Fig. 6(b), the concentration gradient, i.e. the gradient in saturation index, is steeper at higher SRF. Because SRF is largely controlled by the amplitude-to-wavelength ratio, a larger value typically corresponds to the presence of more deep troughs. Consequently, thick hydrodynamic boundary layers and thus larger low velocity zones develop locally (Fig. 6(h)). Such immobile zones cannot be captured by the 2D simulations that do not resolve the geometry explicitly.

The immobile zones can also be a result of the development of recirculation zones at high velocities, when the inertial effects are strong (Bolster et al., 2009, 2014; Cardenas et al., 2009; Bouquain et al., 2012; Sund et al., 2015). At 10 cm/s, the recirculation zones in deep troughs are evident on the vector plot (Fig. 7(a)–(f)). Accordingly, the velocity profile across the fracture aperture shows a switch of the flow direction (Fig. 7(g)). The development of recirculation zones provides another mechanism that traps the solute in the deep troughs (Bolster et al., 2009, 2014; Cardenas et al., 2009; Bouquain et al., 2012; Sund et al., 2015). As a result, high concentrations and low thermodynamic driving force in these regions are reinforced (Fig. 7(a)–(f)). This limits the contribution of surface area in the troughs to the overall reaction rate further. This explains the over-prediction by the 2D simulations and shows that the rough geometry needs to be resolved explicitly to capture the reaction rate. The development of the immobile zones is consistent with a previous study, which showed that an effective surface needs to be placed farther away from the troughs of the rough surface (Guo et al., 2016). In that study, the authors proposed to use a homogeneous effective surface where the no-slip boundary condition is reproduced to replace the rough physical boundary.

It is in regime III where the effects of surface roughness are most significant. As noted by Li et al. (2008), at high flow rates the well-mixed assumption is applicable for flat-surface geometries as rates reach full surface control. However, surface roughness changes this observation by altering the flow and transport processes in the near surface region.

To summarize, the impacts of surface roughness on reaction rate and their quantification can be delineated into three regimes, as highlighted by the color-coded oval shades in Figs. 5(a) and 6(a). Fig. 8 illustrates the three regimes and the corresponding SRF, flow velocity and residence time. In general, at low flow velocities and with long residence time, a large portion of the fluid in the fracture is close to or at equilibrium and the impacts of surface roughness on reaction rate are limited. The increase in reaction rate due to surface area increase can be captured by a well-mixed reactor model that accounts for surface area increase using SRF. At higher flow velocities and thus with shorter residence time, the increase in reaction rate with respect to SRF is more noticeable. The contribution of surface area to reaction rate becomes influenced by longitudinal and transverse concentration gradients. At low roughness, the impacts of surface roughness can be evaluated by accounting for longitudinal and transverse gradients in addition to surface area increase using a 1D or 2D model. At high roughness, large immobile zones due to the presence of a hydrodynamic boundary layer and the recirculation zones develop in deep troughs, resulting in steep transverse concentration gradients. To capture the extent of reaction rate increase due to roughness under these conditions, the rough geometry needs to be resolved explicitly.

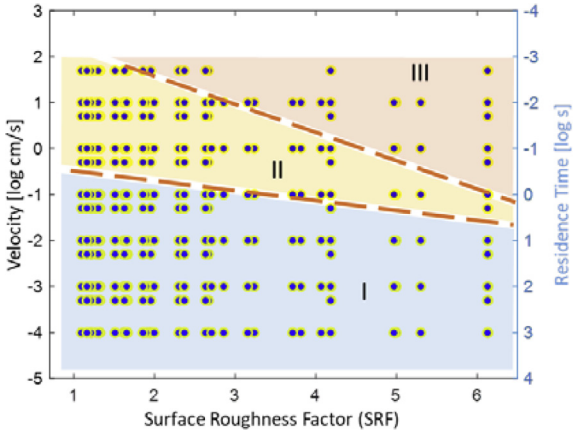


Fig. 8. Data points of the full length fracture simulations (black dots with yellow edge) plotted against surface roughness factor and average velocity [cm/s] and residence time [s]. The shaded areas illustrate the three regimes regarding the impacts of surface roughness on reaction rates with the same color code as Figs. 5 and 6.

3.2. Upscaling of surface roughness

In this subsection, based on the numerical simulation results, we devise a mathematical relationship to capture the compound effects arising from surface roughness on reaction rates.

Non-linear analysis showed that the correction term in Eq. (17) takes the form of a power-law function of SRF.

$$F = \alpha SRF^{\beta} \tag{19}$$

The multiplying factor (α) accounts for the flow-reaction interplay that is missing in the well-mixed reactor model, and the exponent (β) aims to capture the observations that the increase in surface area does not translate into reaction rate increase linearly (Fig. 4). Studies that compare well-mixed reactor simulations with 2D simulations of

smooth fractures showed that the dynamics that are missing in the well-mixed reactor model include longitudinal and transverse diffusion driven by concentration gradients in the fractures (Li et al., 2008). The contribution of surface area increase to reaction rate increase is also affected by local concentration gradients caused by the interplay between local reaction, flow and transport, as shown in the previous section.

Péclet number and Damköhler number have typically been used to indicate the presence and the extent of concentration gradient, and to quantify the resulting modification of macro-transport properties. For example, previous studies have shown that the dispersion coefficients for conservative solute transport in fractures with variable apertures are a function of Péclet number (Bouquain et al., 2012; Richmond et al., 2013), and the macro-transport (and reaction) properties of reacting solutes in periodic pores or channels are dependent on both Péclet number and Damköhler number (Dykaar and Kitanidis, 1996).

Here, the multiplying factor (α) is formulated as an exponential function of Pe and Da_I (Eq. (20)), and falls between 0 and 1. This is consistent with the observation that local concentration gradients tend to result in reaction rates that are lower than those predicted by the well-mixed reactor (Li et al., 2008). Similar formulations have been used in upscaling reaction rate in porous media where concentration gradients arise from heterogeneity in the flow or permeability fields (Soulaine et al., 2017; Wen and Li, 2017).

$$\alpha = 1 - \exp\left(-Pe^{c_1} Da_I^{c_2}\right) \tag{20}$$

Similar to the multiplying factor, the exponent is formulated as a function of Pe and Da_I (Eq. (21)). The exponent is between 0 and 1, and captures the observation that the increase in the reaction rate is less than the increase in surface area across all flow velocities tested.

$$\beta = 1 - \exp\left(-Pe^{c_3} Da_I^{c_4}\right) \tag{21}$$

The coefficients were evaluated using the 2528 data points from the pore-scale reactive transport simulations. Overall, the goodness-of-fitting as measured by the adjusted R^2 is 0.91, which supports the formulation proposed in this study and the physical reasoning behind it. The fitted coefficients c_1 and c_2 are 0.23 and 0.61, respectively, and c_3 and c_4 are 0.19 and 0.58 respectively. The numerical values for c_1, c_3 and c_2, c_4 are similar (95% confidence intervals c_1 = [0.21, 0.25], c_3 = [0.17, 0.2], c_2 = [0.58, 0.64], and c_4 = [0.54, 0.61]). In fact, assuming the same coefficients for α and β results in the same goodness-of-fitting, with coefficients $c_1 = c_3$ of 0.21 and $c_2 = c_4$ of 0.58. These observations indicate that the mechanisms affecting α and β are similar. In other words, the breakdown of the well-mixed reactor assumption (α) is related to the breakdown of the assumption of proportionality between rate and SRF (β). Surface roughness, both via surface area increase and changes in local hydrodynamics, impact the reaction rate by modifying local concentration gradients. This also means that for the conditions tested in this study, the correction term can be simplified, reducing the number of coefficients to two.

This mathematical relationship proposed also captures the three regimes discussed in the previous section. Fig. 9 shows α and β as a function of Pe and Da_I, Eqs. (20) and (21). Both parameters approach one at large Da_I values. The plateau on the parameter surface corresponds to regime I discussed above: when the reaction is substantially faster than the fluid flow, the bulk reaction rate is the reaction rate predicted by the well-mixed reactor model that corrects for the increase in surface area. At lower Da_I values, α and β decrease, indicating the presence of concentration gradients that arise from the hydrodynamic boundary layer and recirculation zones. Surface area increase in the resulting immobile zones does not contribute to the reaction, and the increase in reaction rate is less than SRF. The well-mixed reactor reaction rate needs to be corrected for flow dynamics as well. These observations are consistent with regime II and III.

The transition between regime I and the other two regimes, as indicated by the downward bend of the surfaces at α and β close to one (Fig. 9), is dependent on Pe. As Pe increases, the transition shifts to lower Da_I values. Although the transition between regime II and III is not explicitly accounted for, the power law relationship allows its consideration implicitly. For example, for a given flow rate and thus a fixed β (<1), a larger SRF value corresponds to a larger reduction in surface area contribution to reactions. Consequently, the increase in reaction rate in a

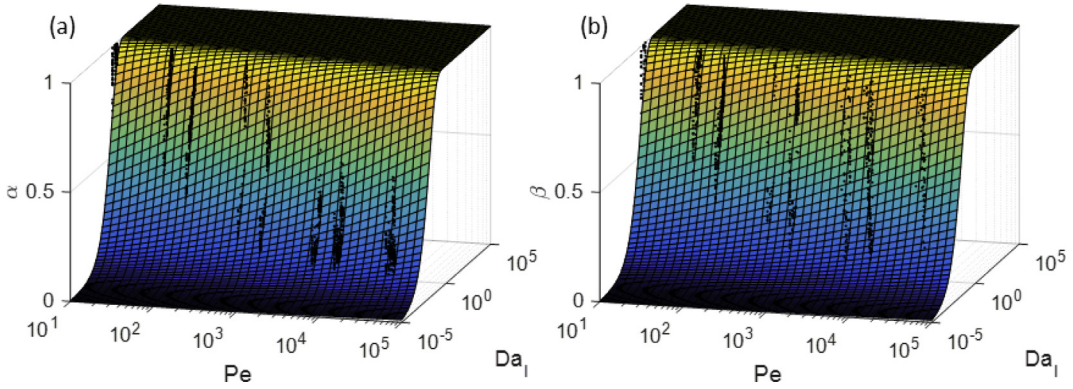


Fig. 9. (a) The multiplying factor (α), and (b) the exponent (β) in relation to Pe and Da_I. The black dots are data points from the pore-scale reactive transport simulations.

rougher fracture is mitigated by a larger extent. This is consistent with the observation that in rougher fractures, recirculation zones are more likely to develop and reaction rate increase deviates from SRF further (Fig. 6(a)).

Some differences are noted between the exponential functions (Eqs. (20) and (21)) in our study and previous studies (Soulaine et al., 2017; Wen and Li, 2017), which may be attributed to the differences in the physical problems investigated, in the definitions of the dimensionless numbers, and in the kinetic treatment of the mineral reactions. Significant similarities in the power law formulation and the strong dependence on the time scales of reaction and transport, however, are observed in all studies. This reiterates the importance of the coupling between reactions and hydrodynamics in controlling the geochemical evolution in different kinds of heterogeneous systems. This also implies that a unified mathematical framework may exist for reaction rate upscaling in heterogeneous systems, which is a research topic that warrants further investigations.

4. DISCUSSION

The proposed relationship provides a useful tool to interpret experimental observations. For example, in the experiments of Levenson et al. (2015), calcareous fractures of 10 mm long and 5 mm wide were exposed to fluid flow at a velocity of about 2 mm/s. The fluid is acidified by HCl and has a pH of 5. The fractures were kept open by a 65 μm spacer, and the SRF ranges between one and two. Analysis of the effluent chemistry showed that the standard deviation of bulk reaction rate is less than 20%. This implies that contribution of surface area increase due to roughness is largely suppressed, which is consistent with what would be expected from the framework proposed in our study. The experimental fractures can be discretized into ten grid cells, such that the size of each grid cell is comparable to the fracture domain used in this study. For each grid cell, the corresponding Pe, Da_I and β values can be calculated to evaluate the extent to which surface area increase contributes to the reaction rate. The simulations with similar SRF and flow rate from our study showed that due to calcite dissolution, pH increases for 2–3 units when the fluid reaches the end of the fracture. Given that the influent pH and geochemical conditions in our study are similar to the experiments, it is expected that the pH dependent k_{rxn} and thus Da_I decreases along the flow direction sharply in the experiemtnal fractures. As a result, β is close to 0.1 for most grid cells, except for the inlet grid cell, which has a value of ~0.6. For a given SRF of 2, the expected increase in reaction rate would be 150% close to the inlet, and 110% in the rest of the fracture. Therefore, the overall reaction rate of a fracture with SRF of 2 is only higher than the reaction rate of a fracture with SRF of 1 by <20%.

While the SRF values used in our study are on the lower end (1–10) of the values reported in the literature, they are nevertheless consistent with values documented for freshly created surfaces (Anbeek, 1992; Hodson, 1998), and in good agreement with the ranges reported for carbonate fractures (Gouze et al., 2003; Noiriel et al., 2007; Levenson et al., 2015). SRF on the order of 100 has also been reported, especially for weathered systems (Anbeek, 1992; White and Brantley, 2003). In those cases, surface area of micro-pores typically accounts for a large portion of the BET surface area, and thus the large SRF values (Anbeek, 1992). Such internal porosity was not considered in this study as the solid phase was assumed to be impermeable. Roughness attributable to micropores, however, is not expected to affect the overall reaction rate (Anbeek, 1992). Previous theoretical study has shown that surface areas within deep pores or pores with narrow throats have negligible contribution to the reactions (Jeschke and Dreybrodt, 2002). As also shown in our study, the bottom of deep troughs, which resembles the flow conditions of micropores, is mostly close to equilibrium. The small local thermodynamic driving force means that the increase in surface area in troughs or due to internal pores will not lead to substantial increase in the overall reaction rate. These observations and the proposed power law formulation also coincide with the scale dependent reaction rate law for basalt weathering developed by Navarre-Sitchler and Brantley (2007), which includes a roughness term with an exponent that is smaller than one, meaning that the contribution of roughness is discounted. The authors also pointed out that the exponent varies with mineralogy, which may be partly due to the differences in fractal structures, and partly due to the differences in reaction rates as indicated by the dependence on Damköhler number shown in this study.

In order to run multiple numerical tests, we limited our simulations to 2D. We note that the 2D structures may have lower surface roughness compared to 3D systems, and flow dynamics may be different (Lee et al., 2017). However, we expect that the findings of this study, including the proposed power law relationship, are still valid because the 2D roughness simulations capture all the fundamental physical processes, i.e. the interplay between surface roughness, reaction rate, and flow conditions. For example, concentration gradients that arise from the hydrodynamic boundary layer and local hydrodynamics such as recirculation zones have also been observed in 3D systems (Molins et al., 2014; Lee et al., 2017). The coefficients, however, may be different, and further investigations are required to quantify the impacts of surface roughness on reaction rate in 3D systems.

The current study focuses on steady state reaction rate and does not consider the dynamic evolution of surface roughness and the subsequent reaction rate change. Nevertheless, the study provides important insights. As shown in Fig. 6(e), the saturation index is closer to zero in the troughs and is more negative at the neighboring peaks. This means that the peaks are likely to be preferentially removed due to the larger thermodynamic driving force for the dissolution reaction. As a result, the roughness is likely to be reduced over time. In presence of mineral heterogeneity, the evolution of roughness may be very different. For example, a highly porous layer can be created by preferential dissolution of fast-reacting minerals in multi-mineral systems (Deng et al., 2016), which creates additional roughness associated with internal porosity. But as discussed above, this type of surface area increase is unlikely to affect the bulk reaction rate substantially due to locally limited flow and thermodynamic driving force.

It should be noted that if the correction term is to be directly applied to upscale reaction rate to scales much larger than what was considered in this study, the formulation of the exponent may need to be revisited. This is primarily because the current formulation involves the use of Da_I, which has some documented limitations such as varying with space and not capturing heterogeneity (Noiriel, 2015). For nanoscale roughness, two caveats are noted, given that roughness below the grid cell size was not captured in the pore-scale simulations. First, this means that the total surface area used to calculate SRF in the analysis is not truly the BET surface area that has been typically used to calculate SRF. The extremely fine scale roughness, however, is not expected to affect processes such as hydrodynamic boundary layer and recirculation, i.e., the overall flow and transport and concentration gradients should be similar. Therefore, like the micrometer roughness and as dictated by the power law formulation, a fraction of the surface area that is associated with the extremely fine roughness will not contribute to the overall reaction because of the small local thermodynamic driving force. Second, the intrinsic reactivity heterogeneity that is associated with nanoscale roughness was not considered. Coupling the impacts of roughness associated with intrinsic reactivity with those of the extrinsic processes investigated in this study would require the resolving of two questions: (1) how the distribution of different reaction sites (e.g. kink sites and face sites), i.e. nanoscale roughness, correlate with micrometer roughness, and (2) how the variations in local reaction rate as a result of heterogeneity in kinetics compare with those caused by changes in thermodynamic driving force due to micrometer roughness related flow and transport processes. Some previous studies, e.g. (Fischer and Luttge, 2007; Bandstra and Brantley, 2015), have examined the interactions between reaction mechanisms and surface topography. Integrating such efforts at molecular scale into pore-scale studies are required to address the aforementioned research questions.

5. CONCLUSIONS

In this study, synthesized 2D rough fracture profiles were used in pore-scale reactive transport modeling to investigate the impacts of surface roughness on reaction rates. The 2D rough profiles include idealized rough geometries and realistic rough profiles that have the same fractal statistical characteristics as natural fractures. The findings across different types of rough profiles are consistent. Increase in surface roughness, as measured by surface roughness factor (SRF), causes the bulk reaction rate of the rough profile to increase. The increase in reaction rate however is not linearly proportional to the increase in SRF; it is dependent on flow and transport conditions. Three regimes were identified. At low flow velocities, the reaction approaches equilibrium and surface roughness does not affect reaction rate substantially. The impacts of surface roughness can be simulated by accounting for surface area increase in a well-mixed reactor. At higher flow velocities, longitudinal and transverse concentration gradients that are affected by surface roughness become important. In order to capture the impacts of surface roughness on reaction rate in these systems, longitudinal and transverse diffusion need to be considered. With low surface roughness, a 1D or 2D model with increased surface area captures the increase in reaction rates due to roughness. As surface roughness increases further, however, strong immobile zones develop because of the hydrodynamic boundary layer and recirculation zones. These processes strengthen local concentration gradients and limit the extent to which the surface area in local troughs contribute to the overall reaction.

These results indicate that fine-scale surface roughness has the potential to extend the range of conditions in which concentration gradients can develop within single pores and fractures and impact rates observed at larger scales. Earlier studies –based exclusively on flat-surface simulations– have suggested that such gradients were only possible in a narrow range of conditions not typically encountered in geological environments (Li et al., 2008).

Our findings also suggest that it is possible to use a well-mixed reactor to simulate the bulk reaction rate in rough fractures, as long as a correction term that accounts for the impacts of surface roughness is included. Based on analysis of the numerical simulations, we proposed a correction term that follows a power law function of SRF. The multiplying factor in the power law relationship, which is an exponential function of Pe and Da_I, corrects for the deviation from the well-mixed assumption under conditions when concentration heterogeneity develops in the fractures. The exponent, which is also formulated as an exponential function of Pe and Da_I and is between 0 and 1, reduces the increase in reaction rate resulting from surface area increase. This is because concentration gradients tend to limit the contribution of surface area in roughness to the reactions. The proposed mathematical relationship provides a useful approach to upscale sub-grid surface roughness in large scale simulations effectively.

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